Canonical GHZ configurations - cluster P2 - helicity conserving

```
species: electron
polarization cluster: P2
source cluster configurations: 97
canonical configurations collected: 10

canonical display criterion: exactly 2 retained final-state components
retained-component cutoff: fraction >= 2.0%
normalized amplitude display: bar length = |A_tilde|, bar color/text = phase

helicity-mechanism subset: helicity conserving
classification: complete-chain R_flip/conserve <= 1
per-page labels: isolated flip/conserve and T/L powers; coherent cross terms are reported separately

cluster retained-component count distribution:
  2 components: 56 configurations
  4 components: 7 configurations
  5 components: 8 configurations
  6 components: 21 configurations
  7 components: 4 configurations
  8 components: 1 configurations

CSV index:
  Output\GradientPhaseSpaceScan\electron\Data\GHZ\dghz\polarization_cluster_02\combined\
  helicity_conserving\canonical_dghz_polarization_cluster_02_helicity_conserving_configu
  rations_electron.csv

Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.
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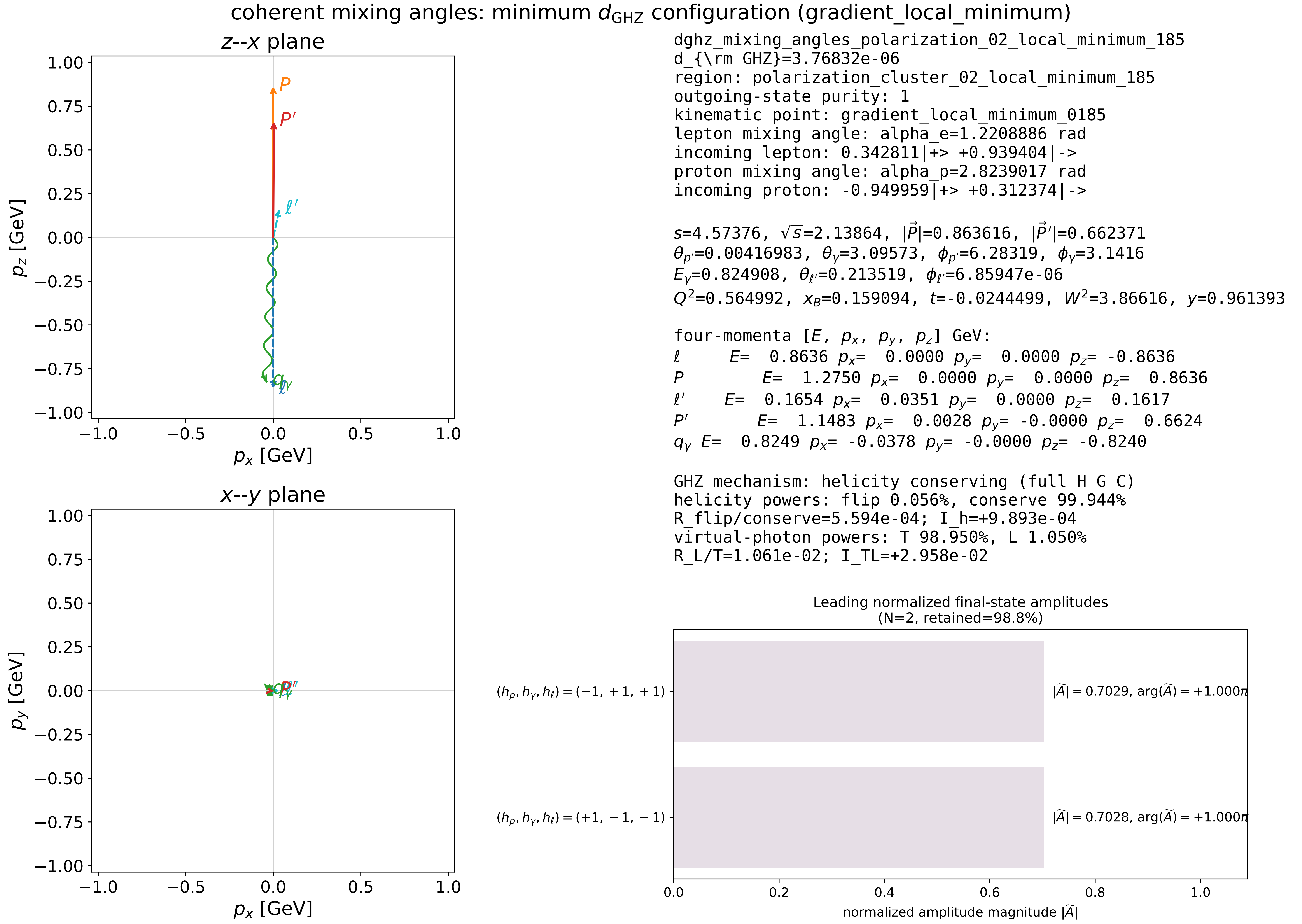
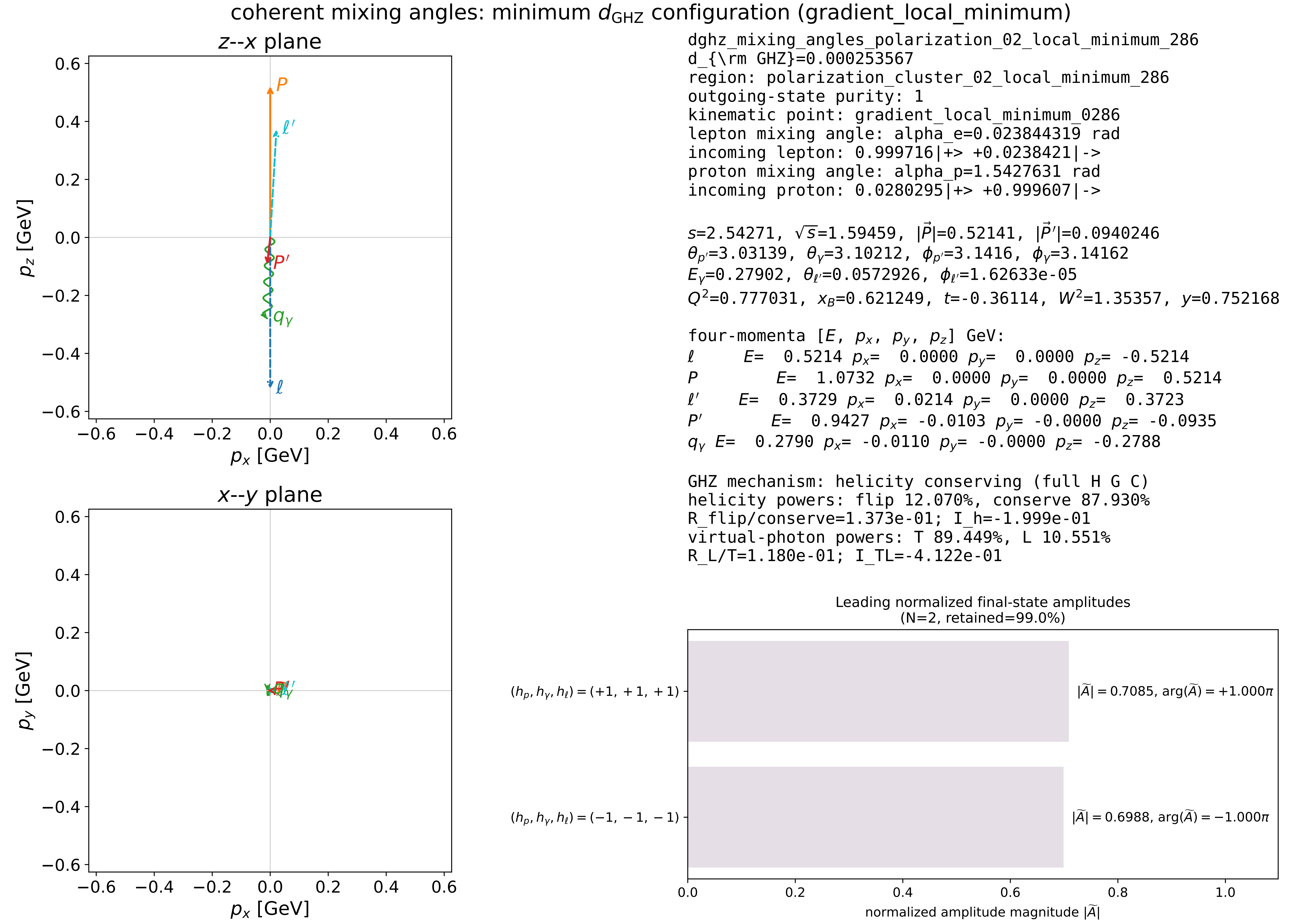
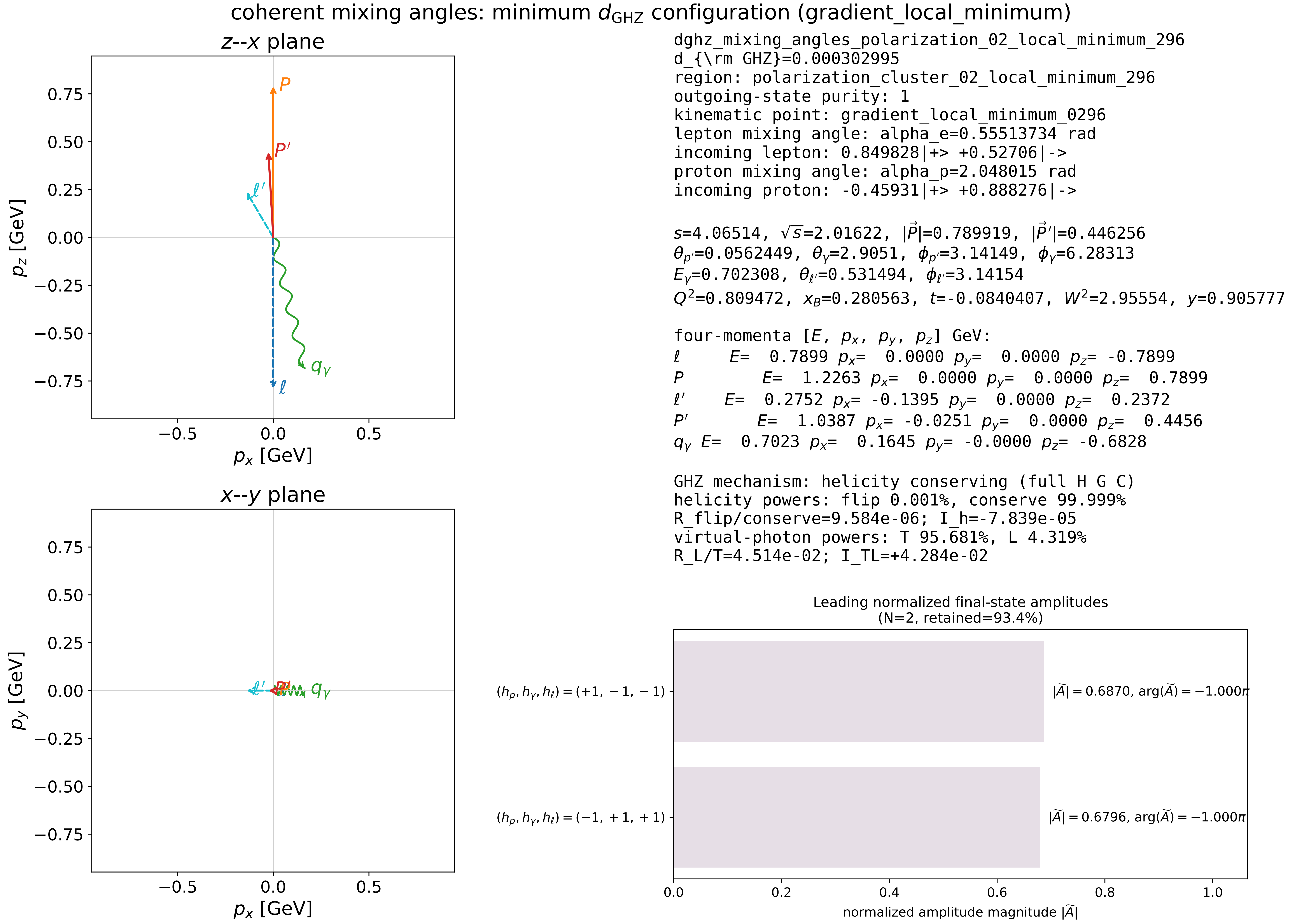
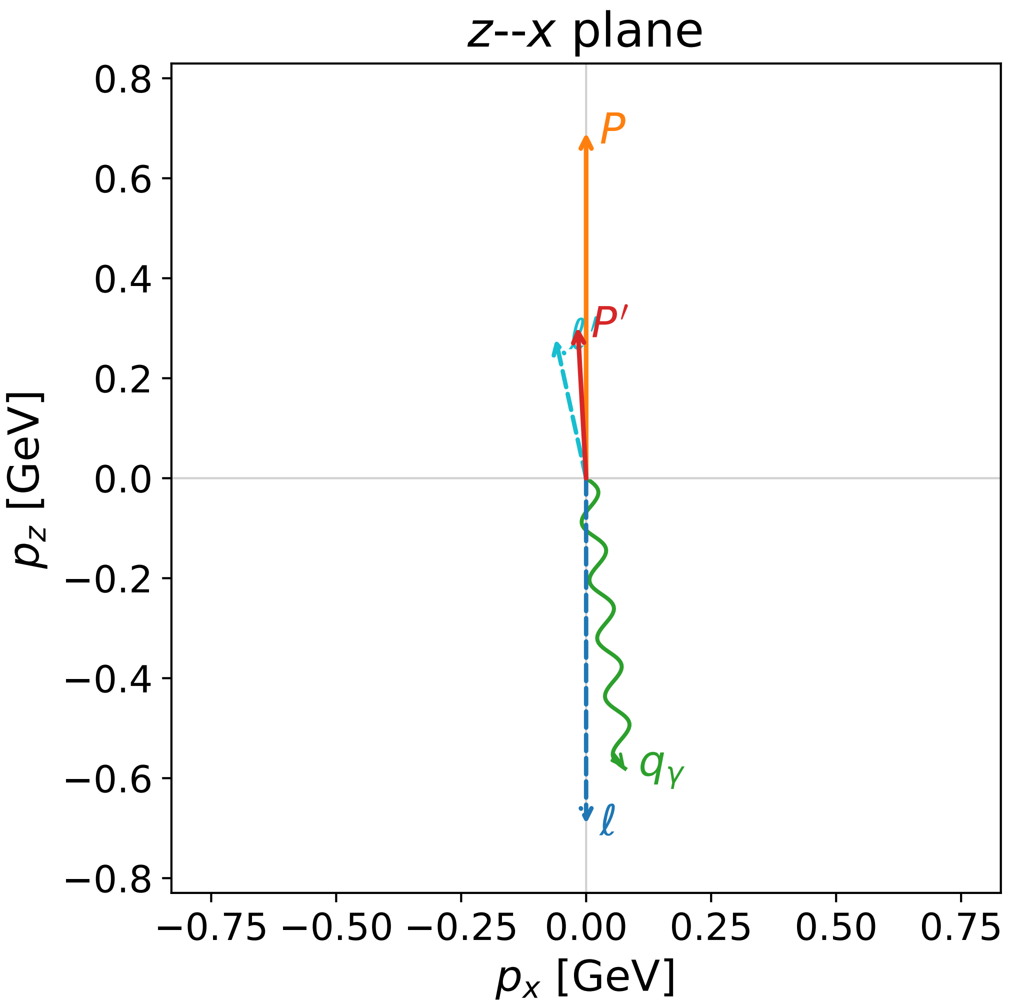
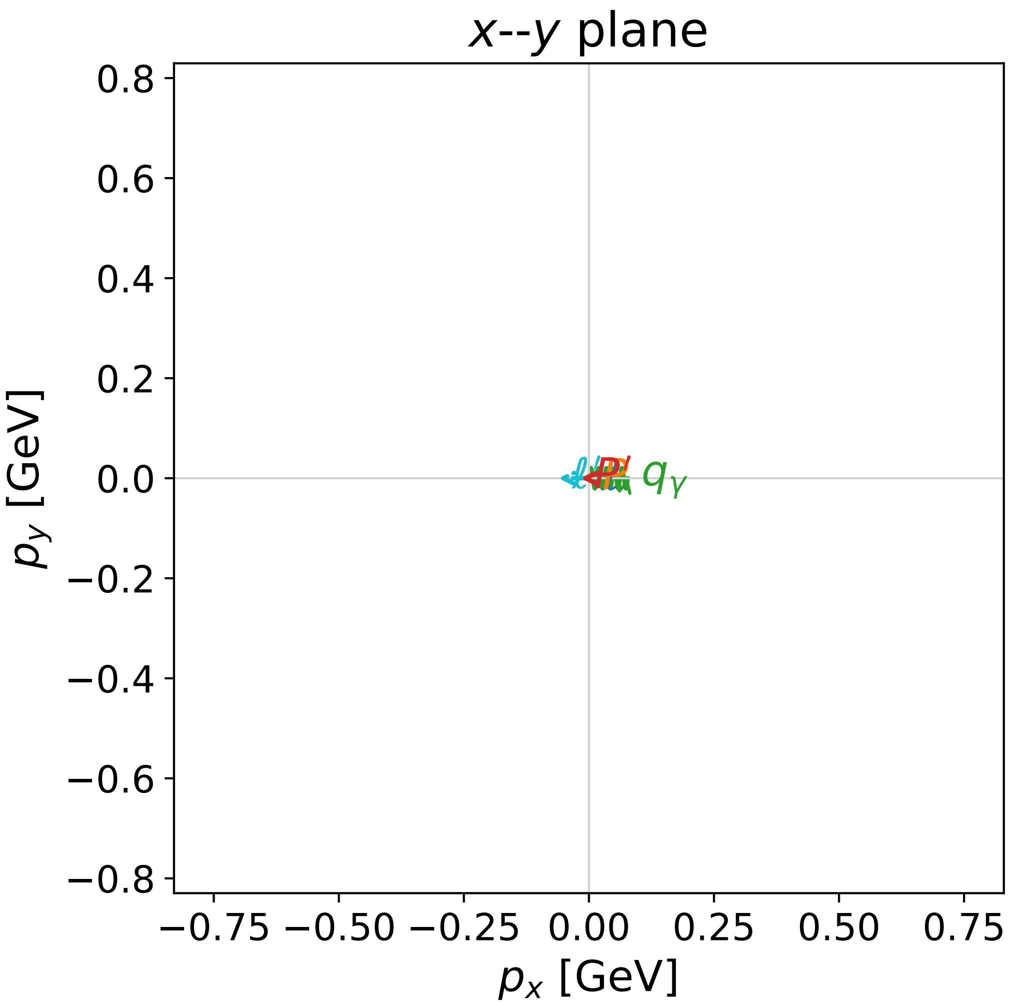

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
z--x plane
P
P'
ℓ'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_185
d_{\rm GHZ}=3.76832e-06
region: polarization_cluster_02_local_minimum_185
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0185
lepton mixing angle: alpha_e=1.2208886 rad
incoming lepton: 0.342811|+> +0.939404|->
proton mixing angle: alpha_p=2.8239017 rad
incoming proton: -0.949959|+> +0.312374|->
s=4.57376, $\sqrt{s}$=2.13864, $|\vec{P}|$=0.863616, $|\vec{P}'|$=0.662371
$\theta_{p'}$=0.00416983, θ_γ=3.09573, $\phi_{p'}$=6.28319, ϕ_γ=3.1416
E_γ=0.824908, $\theta_{\ell'}$=0.213519, $\phi_{\ell'}$=6.85947e-06
Q^2=0.564992, x_B=0.159094, t=-0.0244499, W^2=3.86616, y=0.961393
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.8636 p_x= 0.0000 p_y= 0.0000 p_z= -0.8636
P E= 1.2750 p_x= 0.0000 p_y= 0.0000 p_z= 0.8636
ℓ' E= 0.1654 p_x= 0.0351 p_y= 0.0000 p_z= 0.1617
P' E= 1.1483 p_x= 0.0028 p_y= -0.0000 p_z= 0.6624
q_γ E= 0.8249 p_x= -0.0378 p_y= -0.0000 p_z= -0.8240
GHZ mechanism: helicity conserving (full H G C)
helicity powers: flip 0.056%, conserve 99.944%
R_flip/conserve=5.594e-04; I_h=+9.893e-04
virtual-photon powers: T 98.950%, L 1.050%
R_L/T=1.061e-02; I_TL=+2.958e-02
Leading normalized final-state amplitudes
(N=2, retained=98.8%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
$|\widetilde{A}| = 0.7029$, $\arg(\widetilde{A}) = +1.000\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.7028$, $\arg(\widetilde{A}) = +1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_286
d_{\rm GHZ}=0.000253567
region: polarization_cluster_02_local_minimum_286
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0286
lepton mixing angle: alpha_e=0.023844319 rad
incoming lepton: 0.999716|+> +0.0238421|->
proton mixing angle: alpha_p=1.5427631 rad
incoming proton: 0.0280295|+> +0.999607|->
s=2.54271, $\sqrt{s}$=1.59459, $|\vec{P}|$=0.52141, $|\vec{P}'|$=0.0940246
$\theta_{p'}$=3.03139, θ_γ=3.10212, $\phi_{p'}$=3.1416, ϕ_γ=3.14162
E_γ=0.27902, $\theta_{\ell'}$=0.0572926, $\phi_{\ell'}$=1.62633e-05
Q^2=0.777031, x_B=0.621249, t=-0.36114, W^2=1.35357, y=0.752168
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.5214 p_x= 0.0000 p_y= 0.0000 p_z= -0.5214
P E= 1.0732 p_x= 0.0000 p_y= 0.0000 p_z= 0.5214
ℓ' E= 0.3729 p_x= 0.0214 p_y= 0.0000 p_z= 0.3723
P' E= 0.9427 p_x= -0.0103 p_y= -0.0000 p_z= -0.0935
q_γ E= 0.2790 p_x= -0.0110 p_y= -0.0000 p_z= -0.2788
GHZ mechanism: helicity conserving (full H G C)
helicity powers: flip 12.070%, conserve 87.930%
R_flip/conserve=1.373e-01; I_h=-1.999e-01
virtual-photon powers: T 89.449%, L 10.551%
R_L/T=1.180e-01; I_TL=-4.122e-01
Leading normalized final-state amplitudes
(N=2, retained=99.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$
$|\widetilde{A}| = 0.7085$, $\arg(\widetilde{A}) = +1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$
$|\widetilde{A}| = 0.6988$, $\arg(\widetilde{A}) = -1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
z--x plane
x--y plane
p_z [GeV]
p_y [GeV]
p_x [GeV]
P
P'
ℓ'
q_γ
ℓ
dghz_mixing_angles_polarization_02_local_minimum_296
d_{\rm GHZ}=0.000302995
region: polarization_cluster_02_local_minimum_296
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0296
lepton mixing angle: alpha_e=0.55513734 rad
incoming lepton: 0.849828|+> +0.52706|->
proton mixing angle: alpha_p=2.048015 rad
incoming proton: -0.45931|+> +0.888276|->
s=4.06514, $\sqrt{s}$=2.01622, $|\vec{P}|$=0.789919, $|\vec{P}'|$=0.446256
$\theta_{p'}$=0.0562449, θ_γ=2.9051, $\phi_{p'}$=3.14149, ϕ_γ=6.28313
E_γ=0.702308, $\theta_{\ell'}$=0.531494, $\phi_{\ell'}$=3.14154
Q^2=0.809472, x_B=0.280563, t=-0.0840407, W^2=2.95554, y=0.905777
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.7899 p_x= 0.0000 p_y= 0.0000 p_z= -0.7899
P E= 1.2263 p_x= 0.0000 p_y= 0.0000 p_z= 0.7899
ℓ' E= 0.2752 p_x= -0.1395 p_y= 0.0000 p_z= 0.2372
P' E= 1.0387 p_x= -0.0251 p_y= 0.0000 p_z= 0.4456
q_γ E= 0.7023 p_x= 0.1645 p_y= -0.0000 p_z= -0.6828
GHZ mechanism: helicity conserving (full H G C)
helicity powers: flip 0.001%, conserve 99.999%
R_flip/conserve=9.584e-06; I_h=-7.839e-05
virtual-photon powers: T 95.681%, L 4.319%
R_L/T=4.514e-02; I_TL=+4.284e-02
Leading normalized final-state amplitudes
(N=2, retained=93.4%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.6870$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
$|\widetilde{A}| = 0.6796$, $\arg(\widetilde{A}) = -1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)



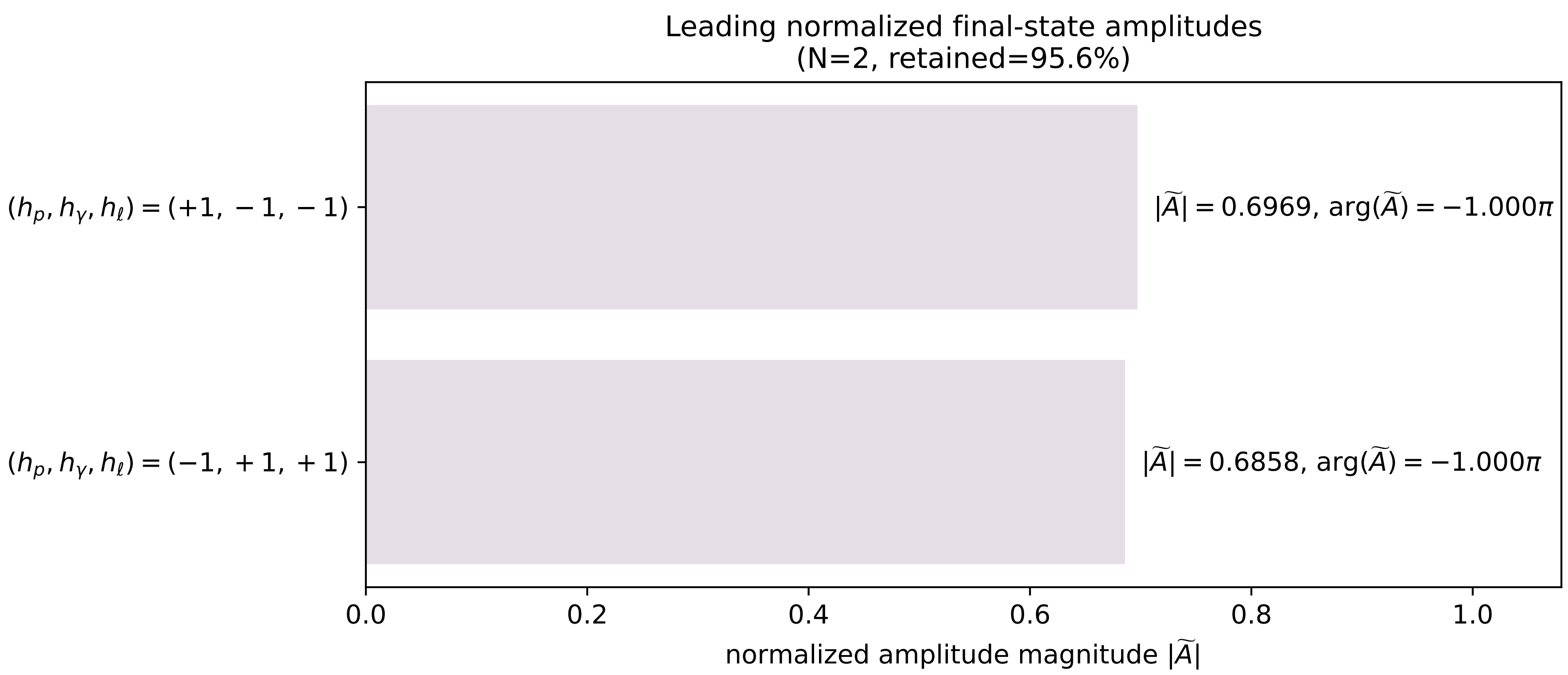


dghz_mixing_angles_polarization_02_local_minimum_308
d_{\rm GHZ}=0.000352127
region: polarization_cluster_02_local_minimum_308
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0308
lepton mixing angle: alpha_e=0.20616489 rad
incoming lepton: 0.978823|+> +0.204708|->
proton mixing angle: alpha_p=1.7420637 rad
incoming proton: -0.170431|+> +0.98537|->

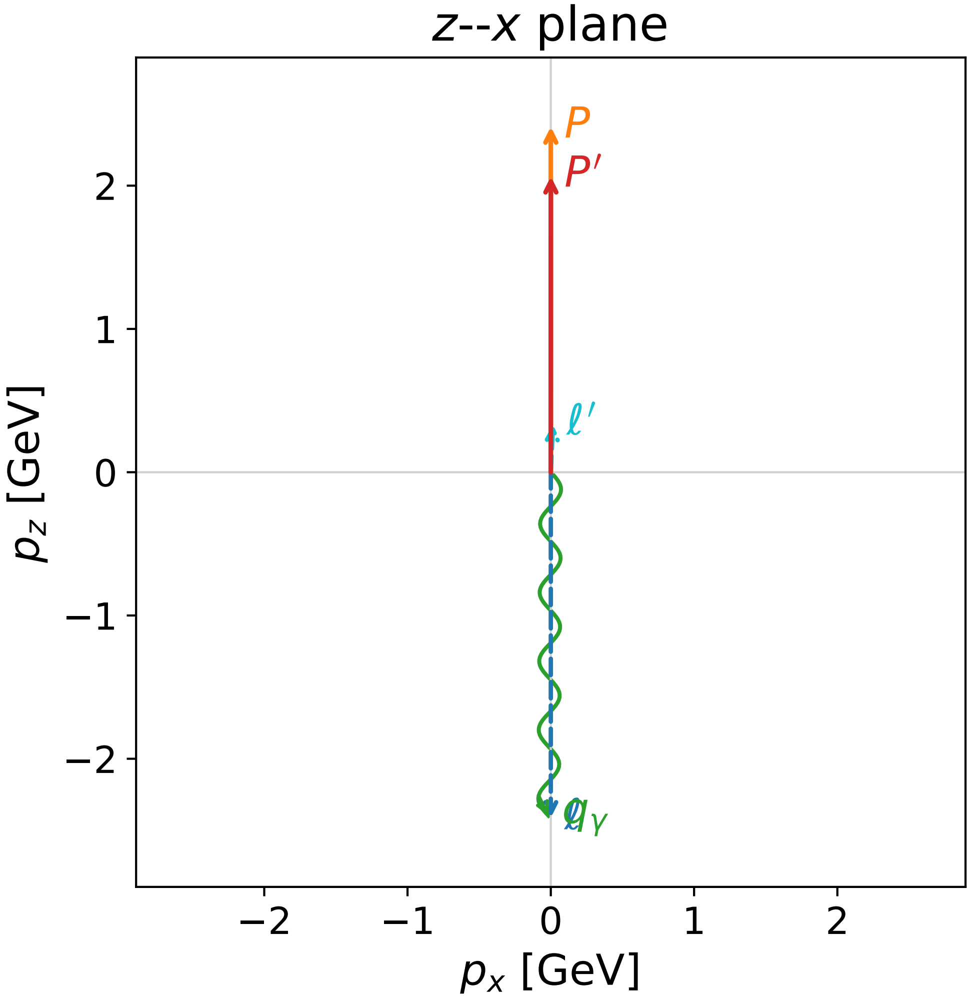
s=3.44654, $\sqrt{s}$=1.85649, $|\vec{P}|$=0.691278, $|\vec{P}'|$=0.302862
$\theta_{p'}$=0.0554148, θ_γ=3.00846, $\phi_{p'}$=3.1414, ϕ_γ=6.28312
E_γ=0.58589, $\theta_{\ell'}$=0.215754, $\phi_{\ell'}$=3.14156
Q^2=0.778686, x_B=0.340408, t=-0.11928, W^2=2.38866, y=0.891225

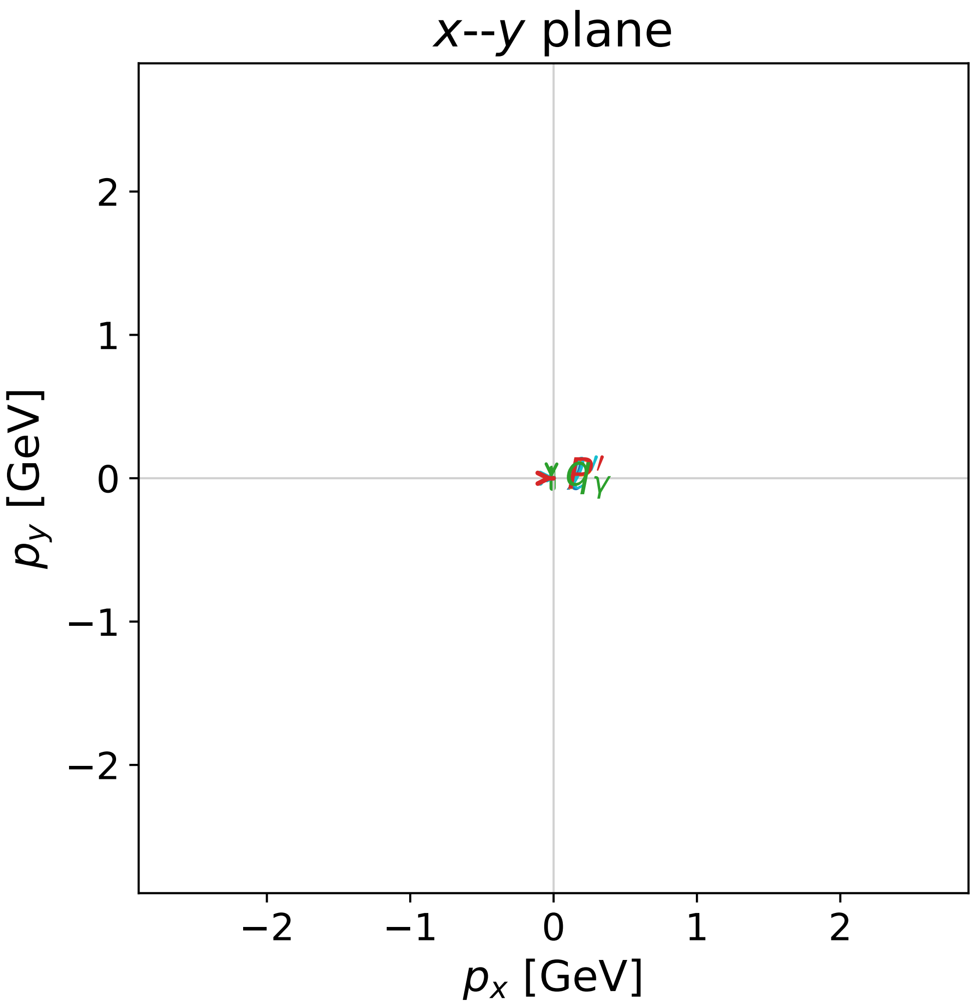
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.6913 p_x= 0.0000 p_y= 0.0000 p_z= -0.6913
P E= 1.1652 p_x= 0.0000 p_y= 0.0000 p_z= 0.6913
ℓ' E= 0.2849 p_x= -0.0610 p_y= 0.0000 p_z= 0.2783
P' E= 0.9857 p_x= -0.0168 p_y= 0.0000 p_z= 0.3024
q_γ E= 0.5859 p_x= 0.0778 p_y= -0.0000 p_z= -0.5807

GHZ mechanism: helicity conserving (full H G C)
helicity powers: flip 0.025%, conserve 99.975%
R_flip/conserve=2.500e-04; I_h=+2.584e-03
virtual-photon powers: T 94.188%, L 5.812%
R_L/T=6.171e-02; I_TL=+6.897e-02



coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)



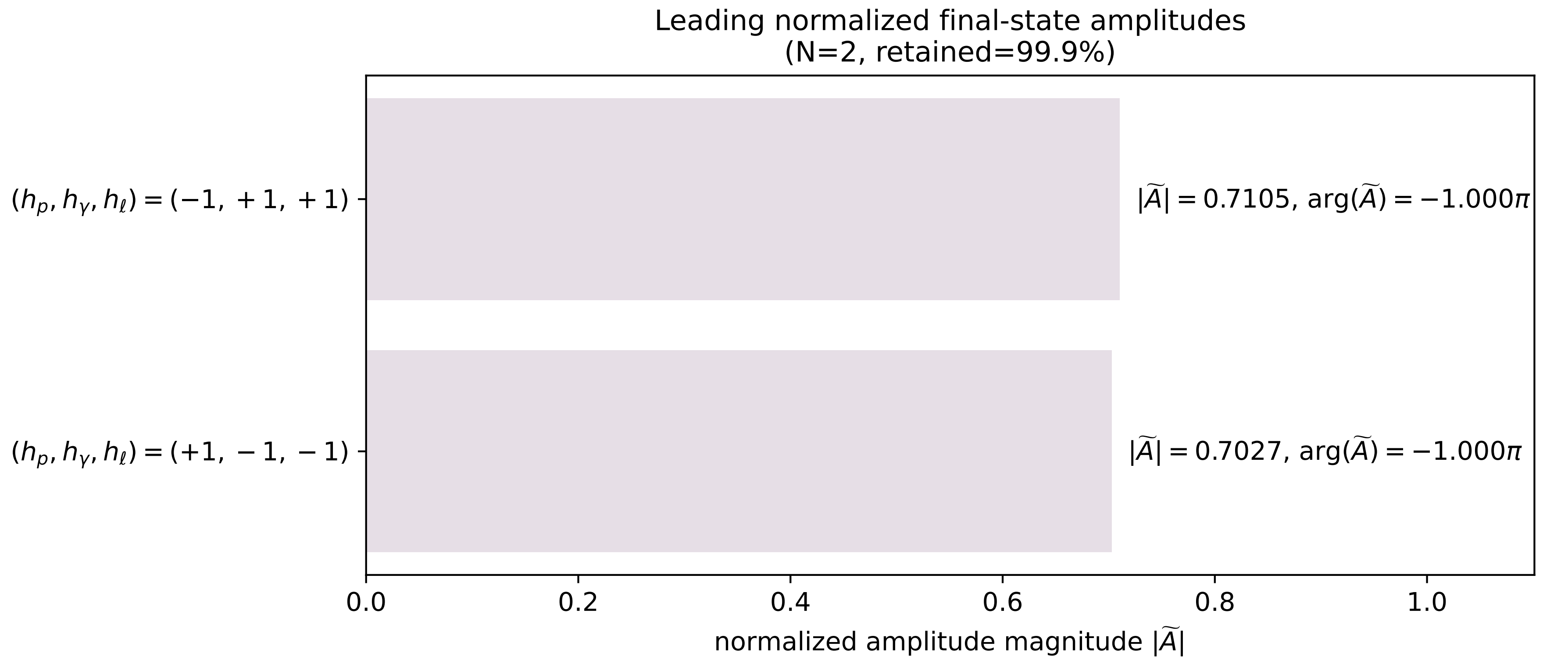


dghz_mixing_angles_polarization_02_local_minimum_323
d_{\rm GHZ}=0.000435487
region: polarization_cluster_02_local_minimum_323
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0323
lepton mixing angle: alpha_e=0.69364314 rad
incoming lepton: 0.768922|+> +0.639343|->
proton mixing angle: alpha_p=2.2770928 rad
incoming proton: -0.649021|+> +0.760771|->

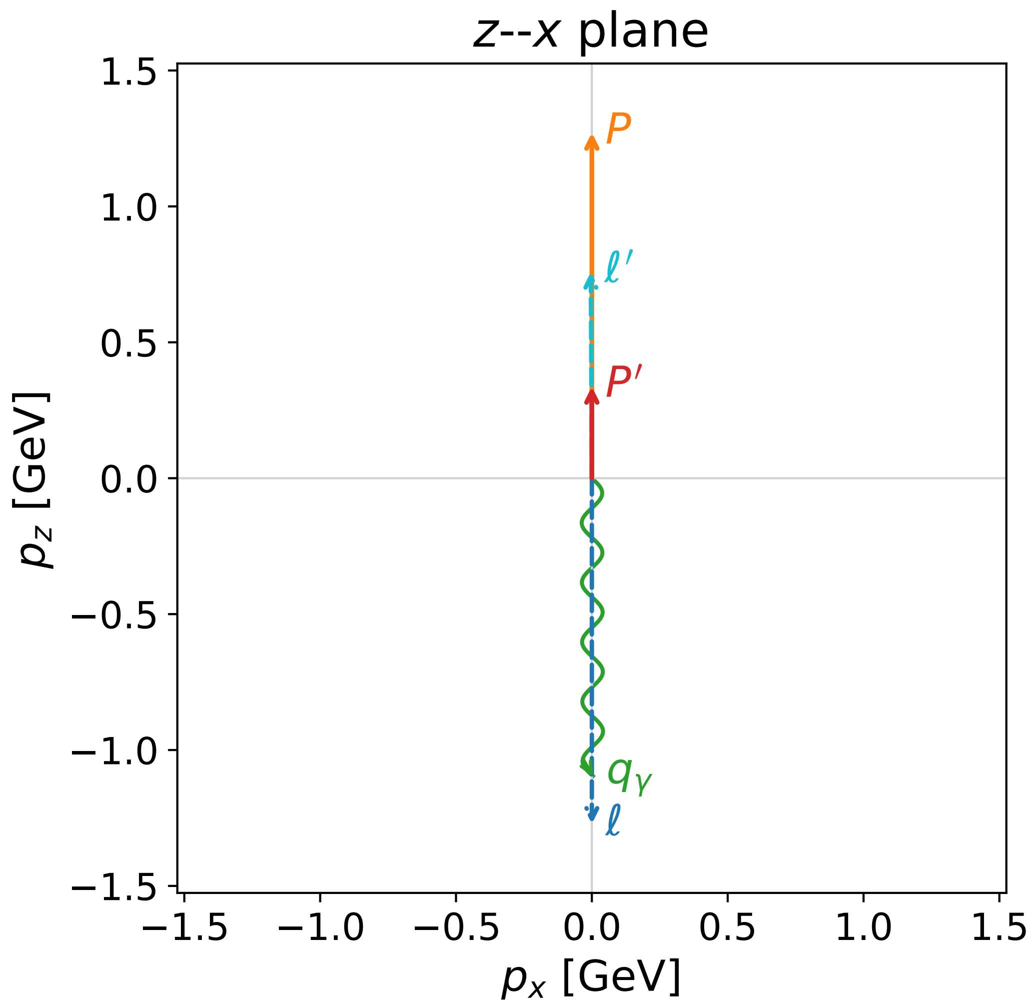
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=2.06443
$\theta_{p'}$=0.000218156, θ_γ=3.13499, $\phi_{p'}$=0.00267229, ϕ_γ=3.14155
E_γ=2.3983, $\theta_{\ell'}$=0.0460371, $\phi_{\ell'}$=6.28306
Q^2=3.22234, x_B=0.134259, t=-0.0181278, W^2=21.6583, y=0.995053

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 0.3342 p_x= 0.0154 p_y= -0.0000 p_z= 0.3338
P' E= 2.2675 p_x= 0.0005 p_y= 0.0000 p_z= 2.0644
q_γ E= 2.3983 p_x= -0.0158 p_y= 0.0000 p_z= -2.3982

GHZ mechanism: helicity conserving (full H G C)
helicity powers: flip 0.109%, conserve 99.891%
R_flip/conserve=1.087e-03; I_h=-6.639e-05
virtual-photon powers: T 99.998%, L 0.002%
R_L/T=2.328e-05; I_TL=-2.146e-04



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

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dghz_mixing_angles_polarization_02_local_minimum_339
d_{\rm GHZ}=0.000560516
region: polarization_cluster_02_local_minimum_339
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0339
lepton mixing angle: alpha_e=1.2628387 rad
incoming lepton: 0.303113|+> +0.952955|->
proton mixing angle: alpha_p=2.8313726 rad
incoming proton: -0.952266|+> +0.305268|->
```
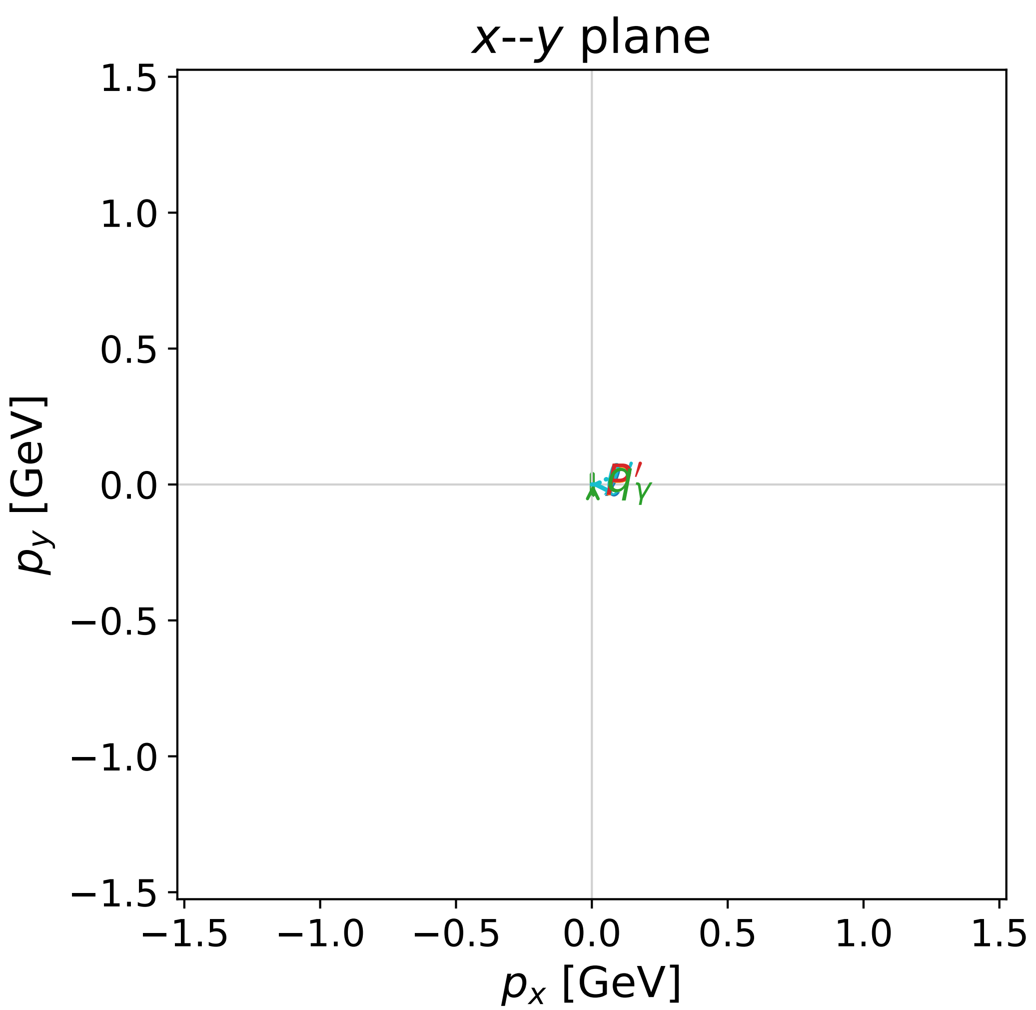
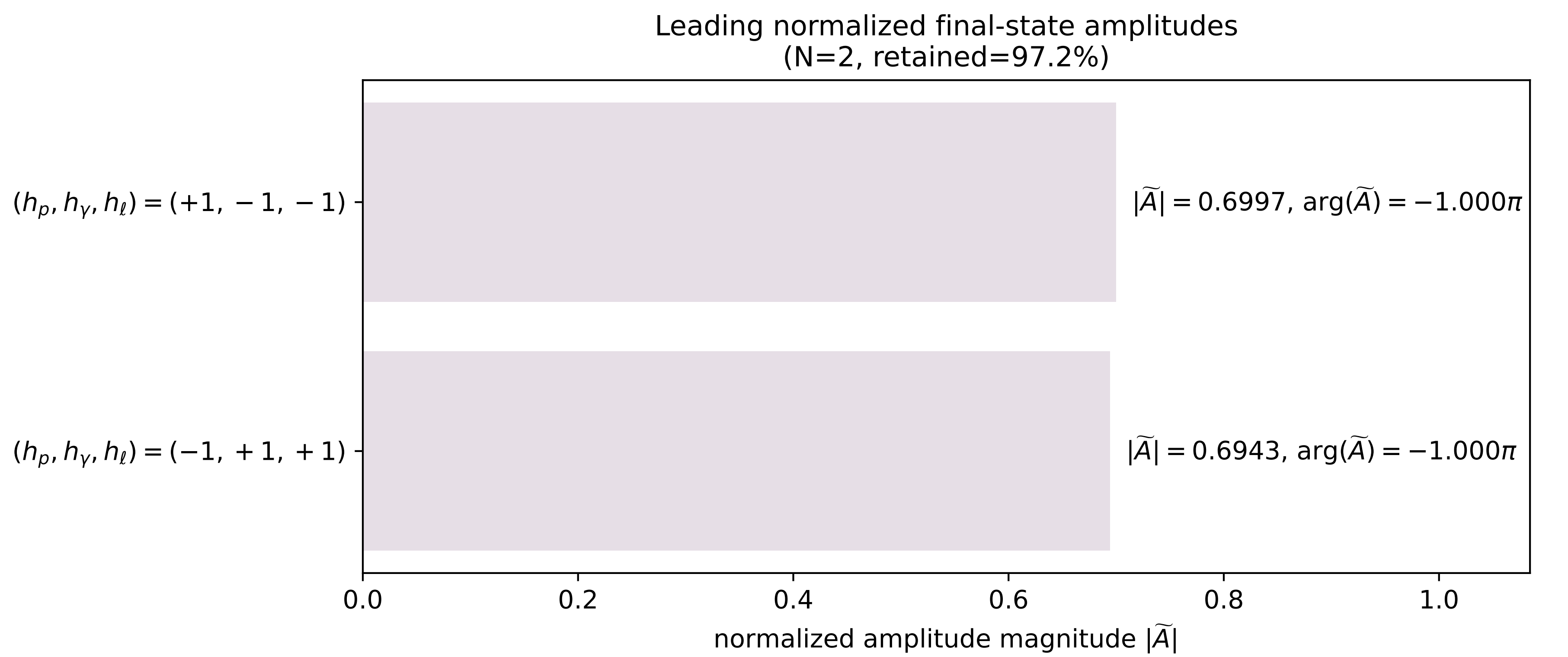

s=8.13048, $\sqrt{s}$=2.8514, $|\vec{P}|$=1.27142, $|\vec{P}'|$=0.337119
$\theta_{p'}$=0, θ_γ=3.13807, $\phi_{p'}$=5.79128, ϕ_γ=2.46636e-05
E_γ=1.09589, $\theta_{\ell'}$=0.005094, $\phi_{\ell'}$=3.14162
Q^2=3.85883, x_B=0.568952, t=-0.532742, W^2=3.80336, y=0.935414

four-momenta [E, p_x, p_y, p_z] GeV:

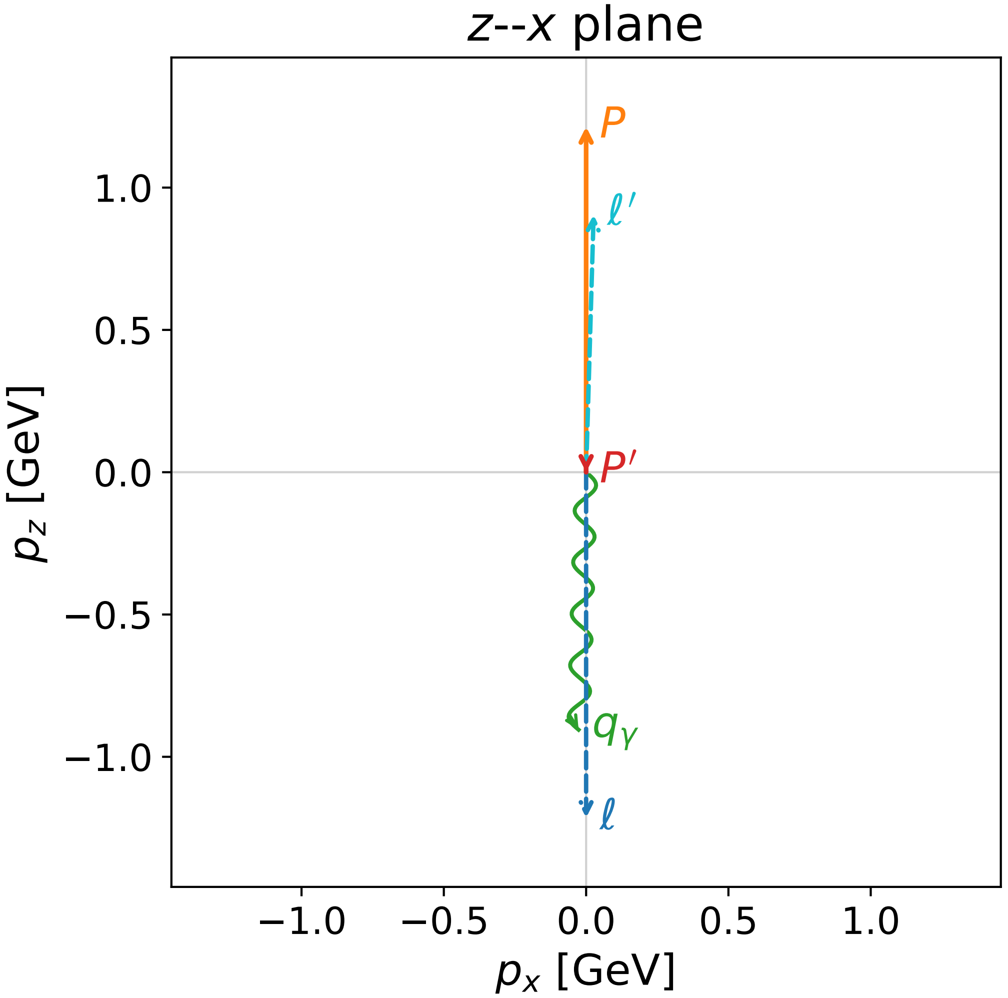
ℓ	E= 1.2714	p_x= 0.0000	p_y= 0.0000	p_z= -1.2714
P	E= 1.5800	p_x= 0.0000	p_y= 0.0000	p_z= 1.2714
ℓ'	E= 0.7588	p_x= -0.0039	p_y= -0.0000	p_z= 0.7588
P'	E= 0.9967	p_x= 0.0000	p_y= -0.0000	p_z= 0.3371
q_γ	E= 1.0959	p_x= 0.0039	p_y= 0.0000	p_z= -1.0959

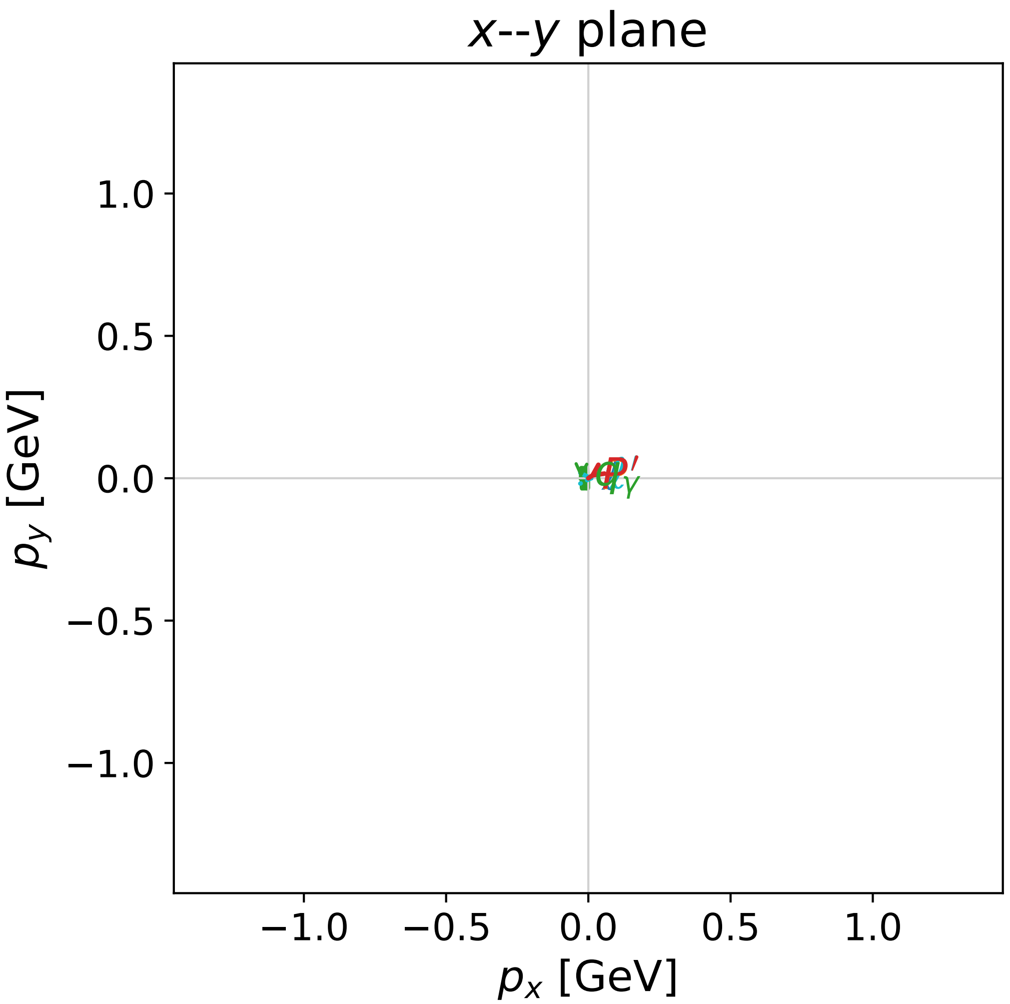
```
GHZ mechanism: helicity conserving (full H G C)
helicity powers: flip 4.483%, conserve 95.517%
R_flip/conserve=4.693e-02; I_h=-3.564e-02
virtual-photon powers: T 100.000%, L 0.000%
R_L/T=2.445e-06; I_TL=+1.962e-04
```





coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)



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dghz_mixing_angles_polarization_02_local_minimum_354
d_{\rm GHZ}=0.0010861
region: polarization_cluster_02_local_minimum_354
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0354
lepton mixing angle: alpha_e=0.027969497 rad
incoming lepton: 0.999609|+> +0.0279659|->
proton mixing angle: alpha_p=1.5433927 rad
incoming proton: 0.0274002|+> +0.999625|->
```

s=7.55541, $\sqrt{s}$=2.74871, $|\vec{P}|$=1.21431, $|\vec{P}'|$=4.5944e-06
$\theta_{p'}$=3.14154, θ_γ=3.11242, $\phi_{p'}$=3.74475, ϕ_γ=3.14156
E_γ=0.905353, $\theta_{\ell'}$=0.0291745, $\phi_{\ell'}$=6.28315
Q^2=4.3966, x_B=0.721341, t=-1.11886, W^2=2.57828, y=0.913036

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.2143 p_x= 0.0000 p_y= 0.0000 p_z= -1.2143
P E= 1.5344 p_x= 0.0000 p_y= 0.0000 p_z= 1.2143
ℓ' E= 0.9054 p_x= 0.0264 p_y= -0.0000 p_z= 0.9050
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= -0.0000
q_γ E= 0.9054 p_x= -0.0264 p_y= 0.0000 p_z= -0.9050

```
GHZ mechanism: helicity conserving (full H G C)
helicity powers: flip 6.745%, conserve 93.255%
R_flip/conserve=7.233e-02; I_h=-8.906e-02
virtual-photon powers: T 99.267%, L 0.733%
R_L/T=7.382e-03; I_TL=-2.961e-02
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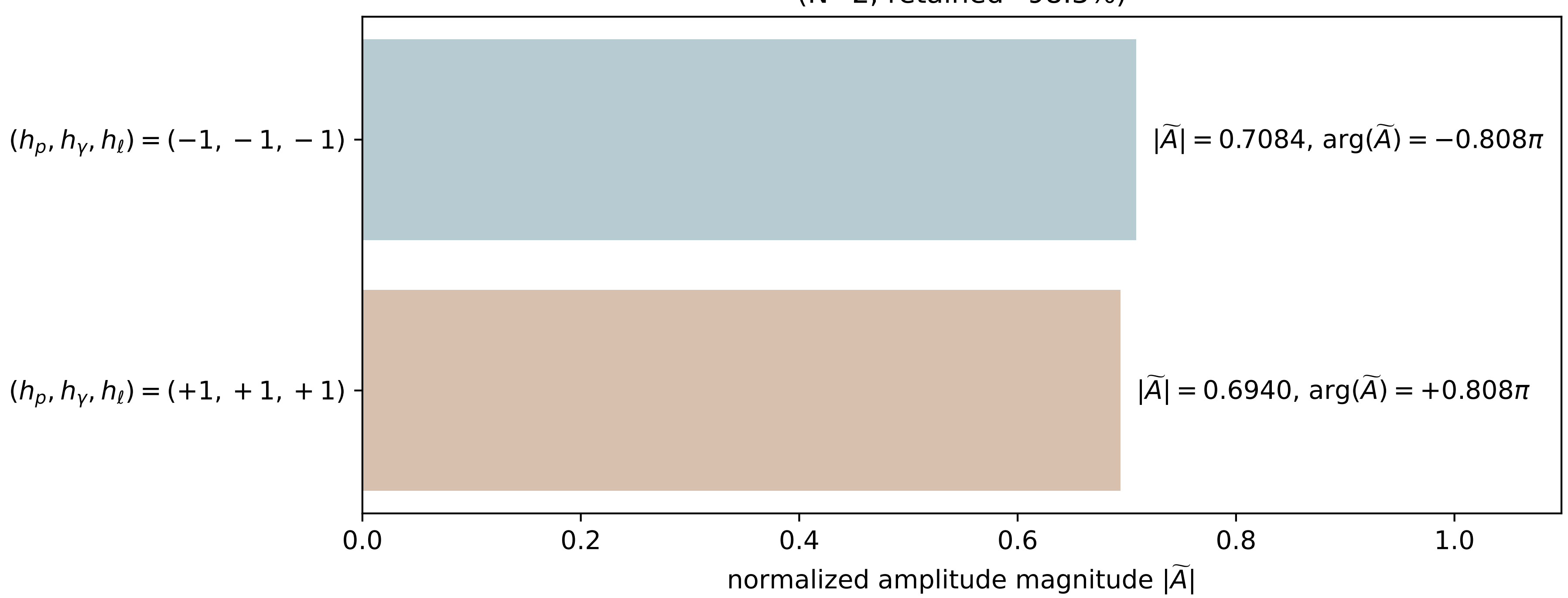
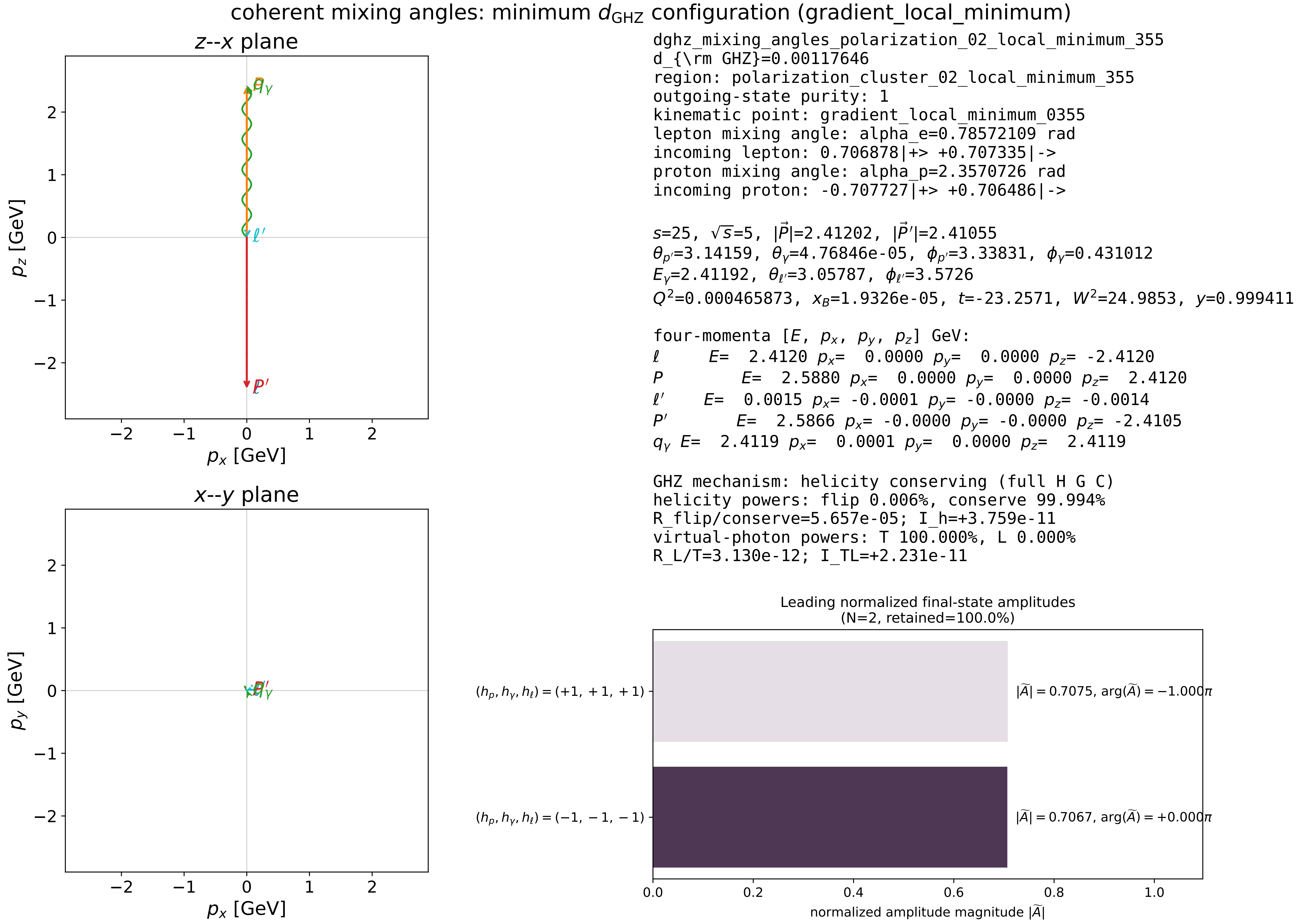
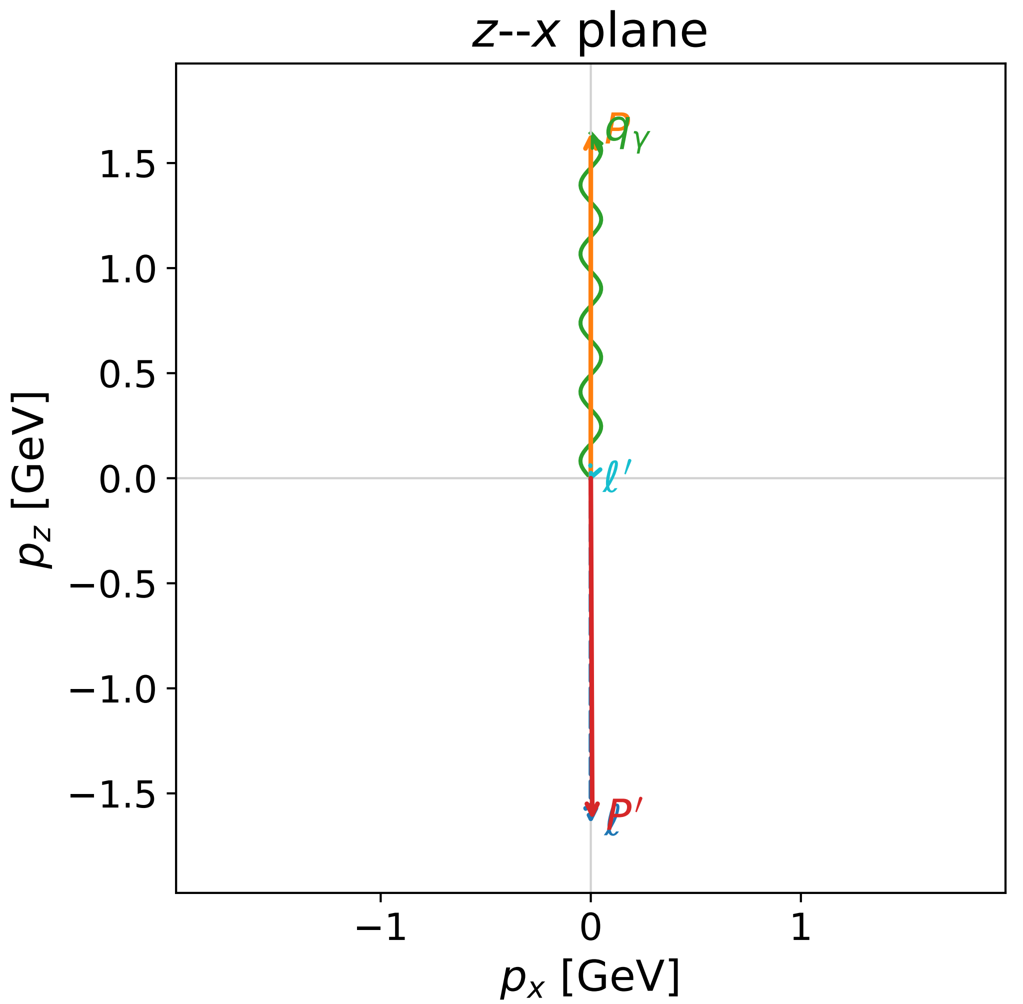
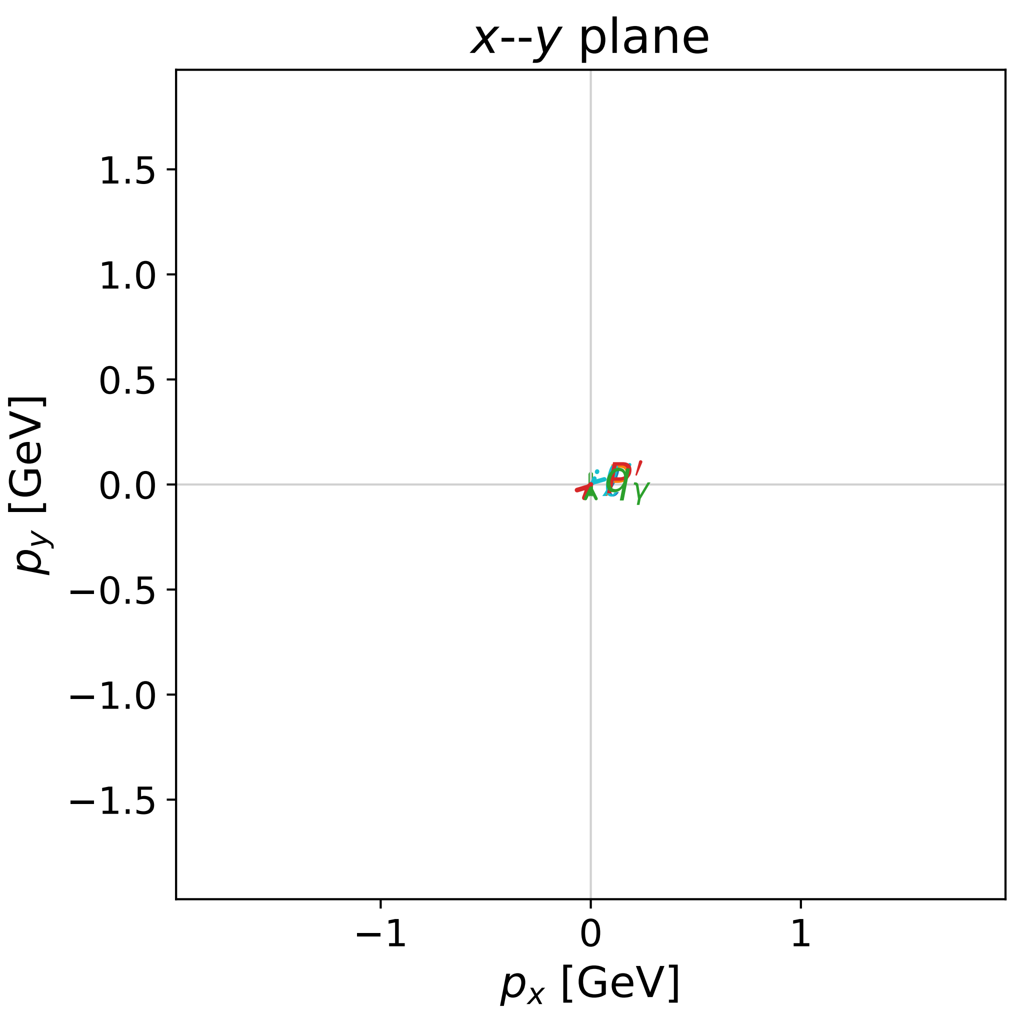
Leading normalized final-state amplitudes
(N=2, retained=98.3%)




coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
q_γ
ℓ'
P'
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_02_local_minimum_355
d_{\rm GHZ}=0.00117646
region: polarization_cluster_02_local_minimum_355
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0355
lepton mixing angle: alpha_e=0.78572109 rad
incoming lepton: 0.706878|+> +0.707335|->
proton mixing angle: alpha_p=2.3570726 rad
incoming proton: -0.707727|+> +0.706486|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=2.41055
$\theta_{p'}$=3.14159, θ_γ=4.76846e-05, $\phi_{p'}$=3.33831, ϕ_γ=0.431012
E_γ=2.41192, $\theta_{\ell'}$=3.05787, $\phi_{\ell'}$=3.5726
Q^2=0.000465873, x_B=1.9326e-05, t=-23.2571, W^2=24.9853, y=0.999411
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 0.0015 p_x= -0.0001 p_y= -0.0000 p_z= -0.0014
P' E= 2.5866 p_x= -0.0000 p_y= -0.0000 p_z= -2.4105
q_γ E= 2.4119 p_x= 0.0001 p_y= 0.0000 p_z= 2.4119
GHZ mechanism: helicity conserving (full H G C)
helicity powers: flip 0.006%, conserve 99.994%
R_flip/conserve=5.657e-05; I_h=+3.759e-11
virtual-photon powers: T 100.000%, L 0.000%
R_L/T=3.130e-12; I_TL=+2.231e-11
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$
$|\widetilde{A}| = 0.7075$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$
$|\widetilde{A}| = 0.7067$, $\arg(\widetilde{A}) = +0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)



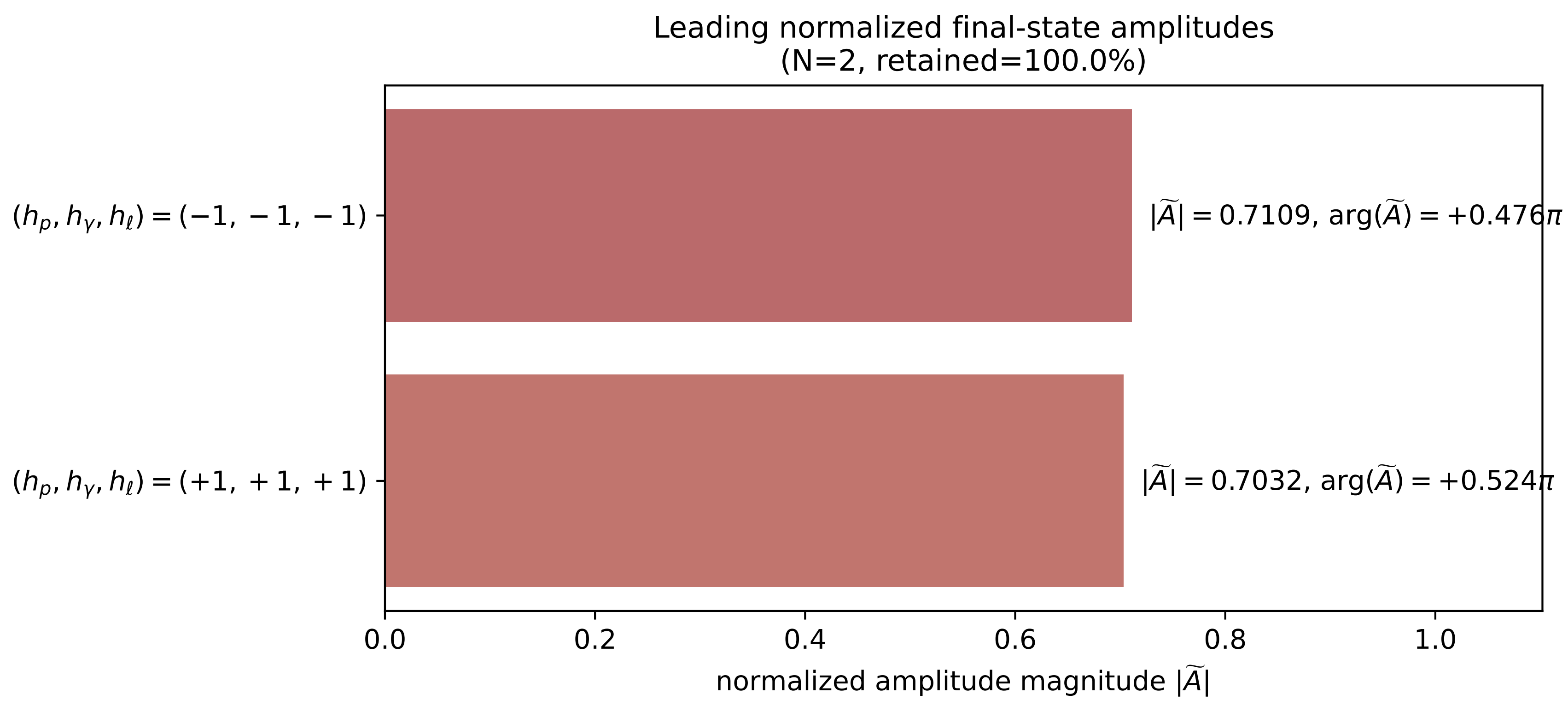


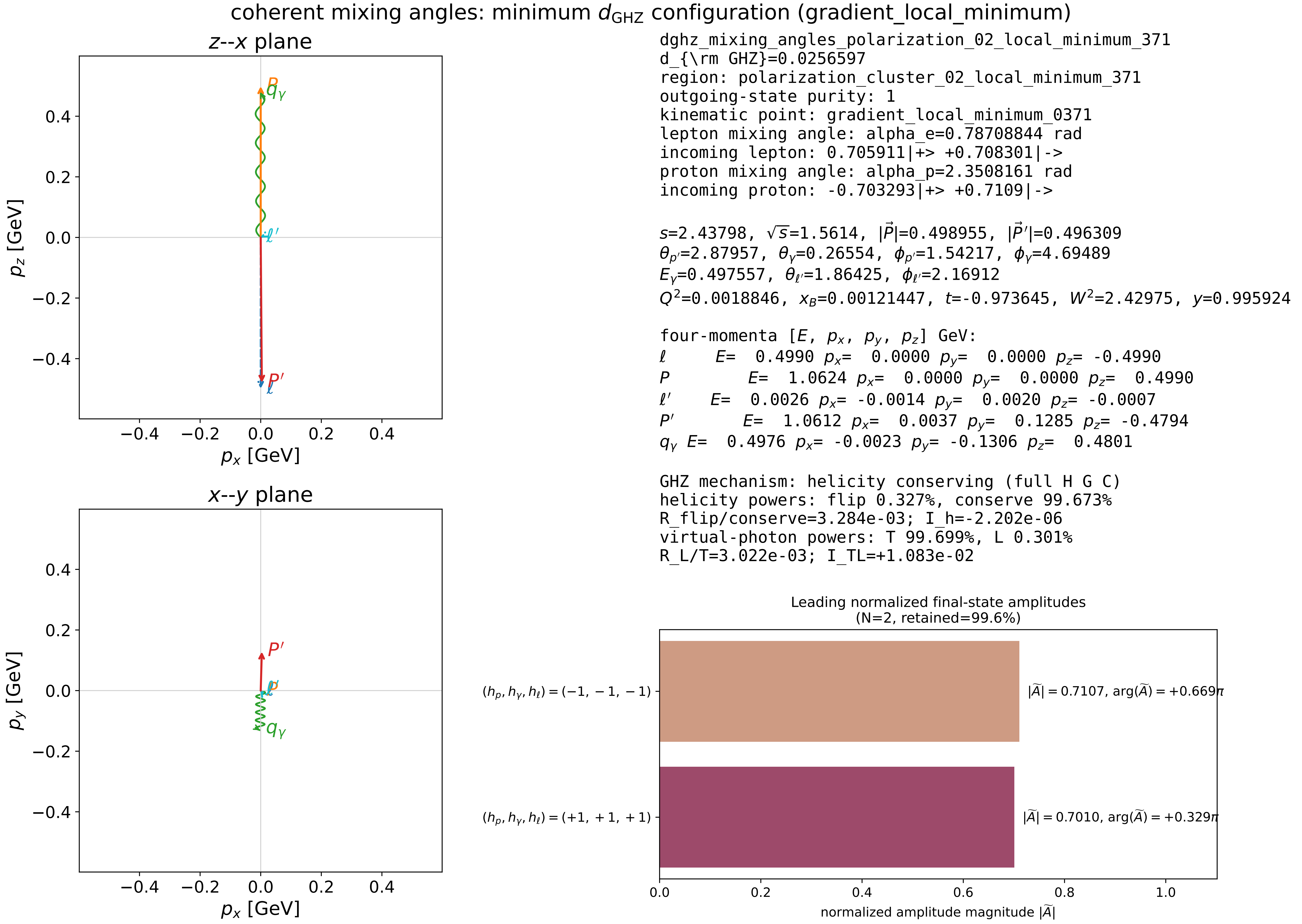
dghz_mixing_angles_polarization_02_local_minimum_370
d_{\rm GHZ}=0.0225057
region: polarization_cluster_02_local_minimum_370
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0370
lepton mixing angle: alpha_e=0.78532657 rad
incoming lepton: 0.707157|+> +0.707056|->
proton mixing angle: alpha_p=2.3500697 rad
incoming proton: -0.702763|+> +0.711424|->

s=12.52, $\sqrt{s}$=3.53836, $|\vec{P}|$=1.64485, $|\vec{P}'|$=1.62509
$\theta_{p'}$=3.13567, θ_γ=0.000106839, $\phi_{p'}$=0.753952, ϕ_γ=0.000369628
E_γ=1.64223, $\theta_{\ell'}$=2.62489, $\phi_{\ell'}$=3.88324
Q^2=0.00850448, x_B=0.000738953, t=-10.6921, W^2=12.3802, y=0.988717

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.6449 p_x= 0.0000 p_y= 0.0000 p_z= -1.6449
P E= 1.8935 p_x= 0.0000 p_y= 0.0000 p_z= 1.6449
ℓ' E= 0.0198 p_x= -0.0072 p_y= -0.0066 p_z= -0.0172
P' E= 1.8764 p_x= 0.0070 p_y= 0.0066 p_z= -1.6251
q_γ E= 1.6422 p_x= 0.0002 p_y= 0.0000 p_z= 1.6422

GHZ mechanism: helicity conserving (full H G C)
helicity powers: flip 0.001%, conserve 99.999%
R_flip/conserve=1.140e-05; I_h=-1.182e-08
virtual-photon powers: T 100.000%, L 0.000%
R_L/T=1.723e-09; I_TL=+7.705e-06




coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
P
q_γ
ℓ'
P'
ℓ
dghz_mixing_angles_polarization_02_local_minimum_371
d_{\rm GHZ}=0.0256597
region: polarization_cluster_02_local_minimum_371
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0371
lepton mixing angle: alpha_e=0.78708844 rad
incoming lepton: 0.705911|+> +0.708301|->
proton mixing angle: alpha_p=2.3508161 rad
incoming proton: -0.703293|+> +0.7109|->
s=2.43798, $\sqrt{s}$=1.5614, $|\vec{P}|$=0.498955, $|\vec{P}'|$=0.496309
$\theta_{p'}$=2.87957, θ_γ=0.26554, $\phi_{p'}$=1.54217, ϕ_γ=4.69489
E_γ=0.497557, $\theta_{\ell'}$=1.86425, $\phi_{\ell'}$=2.16912
Q^2=0.0018846, x_B=0.00121447, t=-0.973645, W^2=2.42975, y=0.995924
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.4990 p_x= 0.0000 p_y= 0.0000 p_z= -0.4990
P E= 1.0624 p_x= 0.0000 p_y= 0.0000 p_z= 0.4990
ℓ' E= 0.0026 p_x= -0.0014 p_y= 0.0020 p_z= -0.0007
P' E= 1.0612 p_x= 0.0037 p_y= 0.1285 p_z= -0.4794
q_γ E= 0.4976 p_x= -0.0023 p_y= -0.1306 p_z= 0.4801
GHZ mechanism: helicity conserving (full H G C)
helicity powers: flip 0.327%, conserve 99.673%
R_flip/conserve=3.284e-03; I_h=-2.202e-06
virtual-photon powers: T 99.699%, L 0.301%
R_L/T=3.022e-03; I_TL=+1.083e-02
Leading normalized final-state amplitudes
(N=2, retained=99.6%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$
$|\widetilde{A}| = 0.7107$, $\arg(\widetilde{A}) = +0.669\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$
$|\widetilde{A}| = 0.7010$, $\arg(\widetilde{A}) = +0.329\pi$
normalized amplitude magnitude $|\widetilde{A}|$